

Newman-Penrose Formalism Calculations

Stephani et al., Equation (35.79)

Metric

$$ds^2 = \frac{2 \left(u^4 W_x(u, x, y)^2 + u^4 W_y(u, x, y)^2 - \sqrt{2} v x \right)}{u^4} du \otimes du - du \otimes dv - \sqrt{2} W_x(u, x, y) du \otimes dx \\ + \sqrt{2} W_y(u, x, y) du \otimes dy - dv \otimes du - \sqrt{2} W_x(u, x, y) dx \otimes du + dx \otimes dx + \sqrt{2} W_y(u, x, y) dy \otimes du + dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(-W_x(u, x, y)^2 - W_y(u, x, y)^2 + \frac{\sqrt{2} v x}{u^4} \right) du + (1) dv + \left(\sqrt{2} W_x(u, x, y) \right) dx + \left(-\sqrt{2} W_y(u, x, y) \right) dy$$

$$m = \left(\sqrt{\frac{1}{2}} \right) dx + \left(i \sqrt{\frac{1}{2}} \right) dy$$

$$\bar{m} = \left(\sqrt{\frac{1}{2}} \right) dx + \left(-i \sqrt{\frac{1}{2}} \right) dy$$

Contravariant components

$$l = (-1) dv$$

$$n = (-1) \, du + \left(-W_x(u, x, y)^2 - W_y(u, x, y)^2 + \frac{\sqrt{2}vx}{u^4} \right) dv$$

$$m = (-W_x(u, x, y) + i W_y(u, x, y)) \, dv + \left(\sqrt{\frac{1}{2}} \right) dx + \left(i \sqrt{\frac{1}{2}} \right) dy$$

$$\overline{m} = (-W_x(u, x, y) - i W_y(u, x, y)) \, dv + \left(\sqrt{\frac{1}{2}} \right) dx + \left(-i \sqrt{\frac{1}{2}} \right) dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u - W_x(u, x, y)^2 - W_y(u, x, y)^2 + \frac{\sqrt{2}vx}{u^4} \partial_v$$

$$\delta = -W_x(u, x, y) + i W_y(u, x, y) \, \partial_v + \sqrt{\frac{1}{2}} \partial_x + i \sqrt{\frac{1}{2}} \partial_y$$

$$\overline{\delta} = -W_x(u, x, y) - i W_y(u, x, y) \, \partial_v + \sqrt{\frac{1}{2}} \partial_x - i \sqrt{\frac{1}{2}} \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{1}{2}i\sqrt{2}\frac{\partial}{\partial y}W_x(u, x, y) + \frac{1}{2}i\sqrt{2}\frac{\partial}{\partial x}W_y(u, x, y)$$

$$\lambda = 0$$

$$\begin{aligned}\nu = & -\sqrt{2}W_x(u, x, y)\frac{\partial}{\partial x}W_x(u, x, y) + i\sqrt{2}W_x(u, x, y)\frac{\partial}{\partial y}W_x(u, x, y) - \sqrt{2}W_y(u, x, y)\frac{\partial}{\partial x}W_y(u, x, y) \\ & + i\sqrt{2}W_y(u, x, y)\frac{\partial}{\partial y}W_y(u, x, y) - \frac{\sqrt{2}xW_x(u, x, y)}{u^4} - \frac{i\sqrt{2}xW_y(u, x, y)}{u^4} + \frac{v}{u^4} - \frac{\partial}{\partial u}W_x(u, x, y) - i\frac{\partial}{\partial u}W_y(u, x, y)\end{aligned}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial y}W_x(u, x, y) + \frac{1}{4}i\sqrt{2}\frac{\partial}{\partial x}W_y(u, x, y) - \frac{\sqrt{2}x}{2u^4}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = 0$$

$$\Phi_{12} = -\frac{1}{2u^4} + \frac{1}{4}i \frac{\partial^2}{\partial x \partial y} W_x(u, x, y) - \frac{1}{4} \frac{\partial^2}{(\partial y)^2} W_x(u, x, y) + \frac{1}{4}i \frac{\partial^2}{(\partial x)^2} W_y(u, x, y) - \frac{1}{4} \frac{\partial^2}{\partial x \partial y} W_y(u, x, y)$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{1}{2u^4} - \frac{1}{4}i \frac{\partial^2}{\partial x \partial y} W_x(u, x, y) - \frac{1}{4} \frac{\partial^2}{(\partial y)^2} W_x(u, x, y) - \frac{1}{4}i \frac{\partial^2}{(\partial x)^2} W_y(u, x, y) - \frac{1}{4} \frac{\partial^2}{\partial x \partial y} W_y(u, x, y)$$

$$\begin{aligned} \Phi_{22} = & -\frac{\partial}{\partial x} W_x(u, x, y)^2 - W_x(u, x, y) \frac{\partial^2}{(\partial x)^2} W_x(u, x, y) - \frac{1}{2} \frac{\partial}{\partial y} W_x(u, x, y)^2 - W_x(u, x, y) \frac{\partial^2}{(\partial y)^2} W_x(u, x, y) + \frac{\partial}{\partial y} W_x(u, x, y) \frac{\partial}{\partial x} W_y(u, x, y) \\ & - \frac{1}{2} \frac{\partial}{\partial x} W_y(u, x, y)^2 - W_y(u, x, y) \frac{\partial^2}{(\partial x)^2} W_y(u, x, y) - \frac{\partial}{\partial y} W_y(u, x, y)^2 - W_y(u, x, y) \frac{\partial^2}{(\partial y)^2} W_y(u, x, y) \\ & - \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial x} W_x(u, x, y) + \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial y} W_y(u, x, y) - \frac{x \frac{\partial}{\partial x} W_x(u, x, y)}{u^4} + \frac{x \frac{\partial}{\partial y} W_y(u, x, y)}{u^4} - \frac{2 W_x(u, x, y)}{u^4} \end{aligned}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = -\frac{1}{2u^4} + \frac{1}{4}i \frac{\partial^2}{\partial x \partial y} W_x(u, x, y) + \frac{1}{4} \frac{\partial^2}{(\partial y)^2} W_x(u, x, y) + \frac{1}{4}i \frac{\partial^2}{(\partial x)^2} W_y(u, x, y) + \frac{1}{4} \frac{\partial^2}{\partial x \partial y} W_y(u, x, y)$$

$$\begin{aligned}
\Psi_4 = & -\frac{\partial}{\partial x} W_x(u, x, y)^2 - W_x(u, x, y) \frac{\partial^2}{(\partial x)^2} W_x(u, x, y) + 2i W_x(u, x, y) \frac{\partial^2}{\partial x \partial y} W_x(u, x, y) + 2i \frac{\partial}{\partial x} W_x(u, x, y) \frac{\partial}{\partial y} W_x(u, x, y) + \frac{\partial}{\partial y} W_x(u, x, y)^2 \\
& + W_x(u, x, y) \frac{\partial^2}{(\partial y)^2} W_x(u, x, y) - \frac{\partial}{\partial x} W_y(u, x, y)^2 - W_y(u, x, y) \frac{\partial^2}{(\partial x)^2} W_y(u, x, y) + 2i W_y(u, x, y) \frac{\partial^2}{\partial x \partial y} W_y(u, x, y) + 2i \frac{\partial}{\partial x} W_y(u, x, y) \frac{\partial}{\partial y} W_y(u, x, y) \\
& + \frac{\partial}{\partial y} W_y(u, x, y)^2 + W_y(u, x, y) \frac{\partial^2}{(\partial y)^2} W_y(u, x, y) - \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial x} W_x(u, x, y) + \frac{1}{2} i \sqrt{2} \frac{\partial^2}{\partial u \partial y} W_x(u, x, y) - \frac{1}{2} i \sqrt{2} \frac{\partial^2}{\partial u \partial x} W_y(u, x, y) \\
& - \frac{1}{2} \sqrt{2} \frac{\partial^2}{\partial u \partial y} W_y(u, x, y) - \frac{x \frac{\partial}{\partial x} W_x(u, x, y)}{u^4} + \frac{i x \frac{\partial}{\partial y} W_x(u, x, y)}{u^4} - \frac{i x \frac{\partial}{\partial x} W_y(u, x, y)}{u^4} - \frac{x \frac{\partial}{\partial y} W_y(u, x, y)}{u^4} - \frac{2 W_x(u, x, y)}{u^4} - \frac{2i W_y(u, x, y)}{u^4}
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "III"